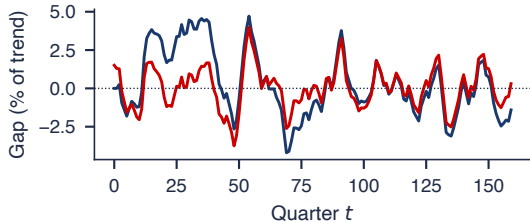
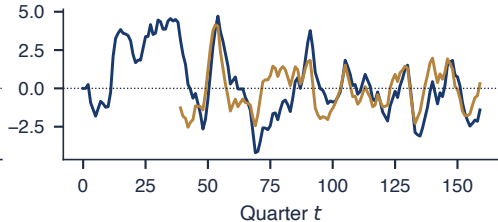


$$100\log \text{GDP}_t = \mu_t + \psi_t$$



The last HP values are pulled to zero



— True cycle ψ_t — HP gap (two-sided, full sample) — HP gap at the end of each sample (real time)